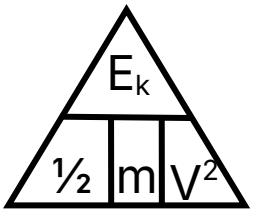
Calculating Kinetic Energy, E_k

The kinetic energy of a moving object is dependent on 2 factors; the mass (m) of the object & its velocity (v). The formula for calculating kinetic energy is:



$E_k = ½ mv^2$

Kinetic energy = ½ x mass x velocity²

- Mass of the object is measured in kg
- Velocity is measured in m/s or ms^{-1} (either one is accepted as they are the same unit of measurement)

Example

The total mass of a go-kart and the driver is 90kg. The go-kart is travelling at 40 m/s, how much kinetic energy does it possess?

What are my knowns?

E_k* = ½ *mv²

= ½ x 90 x 40²

= 0.5 x 90 x 1600

= 72,000 J

= 72 kJ



Questions

(Calculate answer to two significant figures)

1. The first driver of a go-kart had a mass of 50kg and travelled a top speed of 40m/s. They were replaced by a driver with a mass of 70kg. Calculate the kinetic energy both drivers in the go-kart possess when travelling at the same speed. Comment on the difference you find and why it occurs.

2. $E_{k\ driver\ 1} = ½\ mv^2$
3. $= ½ \times 50 \times 40^2$
4. $= 0.5 \times 50 \times 1600$
5. $= 50{,}000$ J
6. **$= 40$ kJ**

$E_{k\ driver\ 2} = ½\ mv^2$

$= ½ \times 70 \times 40^2$

$= 0.5 \times 70 \times 1600$

$= 56{,}000$ J

$= 56$ kJ

The second driver has more kinetic energy than the first driver as their mass is larger.

1. A crane raises a load of 200kg to a height of 30m in one minute. Determine the kinetic energy of the load when it's moving.

$E_k = ½\ mv^2$

$= ½ \times 200 \times 0.5^2$

$= 25$ J

$v = distance / time = d/t = 30 / 60 = 0.5 ms^{-1}$

2. A racing car drives around a circuit at a constant speed of 50m/s. If the car has a kinetic energy of 500kJ, what is the mass of the car?

$E_k = ½\ mv^2$

$m = 2E_k / v^2$

$= 2 \times 500{,}000 / 50^2$

$= 0.5 \times 70 \times 1600$

$= 400$kg

3. A plane has a mass of 9000kg. It flies at a speed of 100m/s. Calculate the kinetic energy of the plane.

   $E_k = ½ mv^2$

   $= ½ \times 9000 \times 100^2$

   $= 45{,}000{,}000$ J

   = 45 MJ

4. A runner crosses the finishing line at a speed of 7m/s. At this point the runner has a kinetic energy of 1862J. Calculate the mass of the runner.

   $E_k = ½ mv^2$

   $m = 2E_k / v^2$

   $= 2 \times 1862 / 7^2$

   = 76kg

5. A remote controlled car has a speed of 1.2m/s. It has a kinetic energy of 0.216J. Calculate the mass of the car.

   $E_k = ½ mv^2$

   $m = 2E_k / v^2$

   $= 2 \times 0.216 / 1.2^2$

   = 0.3kg

6. A lorry of mass 2800kg is travelling at a constant speed. It has a kinetic energy of 132kJ. Calculate the speed of the lorry.

   $E_k = ½ mv^2$

   $v^2 = 2E_k / m$

   =

   = 9.710083ms^{-1}

   = 9.7ms^{-1} to 2 significant figures

7. A motorbike has a mass of 250kg and is travelling at 18m/s. A car has a mass of 950kg and travels at 12.5m/s. Calculate which has the greater kinetic energy.

Motorbike: $E_k = ½ mv^2$

$= ½ \times 250 \times 18^2$

$= 40{,}500$ J = **41,000J** to 2 significant figures

$= 40.5$ MJ **= 41kJ** to 2 significant figures

Car: $E_k = ½ mv^2$

$= ½ \times 950 \times 12.5^2$

$= 74{,}218.75$ J = **74,000J** to 2 significant figures

= 74kJ to 2 significant figures

8. A car of mass 850kg is travelling at a speed of 20m/s. The brakes are applied and the car reduces to a speed of 12m/s. Calculate the change is the car's kinetic energy.

Car start: $E_{k1} = ½ mv^2$

$= ½ \times 850 \times 20^2$

$= 170{,}000$ J

Car end: $E_{k2} = ½ mv^2$

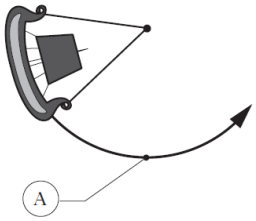
$= ½ \times 850 \times 12^2$

$= 61{,}200$ J

Change in car's kinetic energy, $\Delta E_k = E_{k1} - E_{k2} = 170{,}000 – 61{,}200 = 108{,}800$ J

= **110,000 J** to two significant figures = 110kJ

9. A fairground swing is shown.



The swing and passengers have a combined mass of 2500kg.

(a) Calculate, showing all working and units:

(i) The kinetic energy at its lowest point when the velocity is 15 m/s;

$E_k = ½ mv^2$

$= ½ \times 2500 \times 15^2$

$= 281,250$ J

=**280,000 J** to two significant figures

(ii) The maximum height the swing will reach. (Assume no loss of energy). Note: You must use the potential energy equation to solve this problem: $E_p=mgh$. The next theory page explains calculating E_p.

If no loss of energy, then potential energy, E_P must equal the kinetic energy, E_k.

$E_k = E_P$

$E_k = mgh$

$= 11.4795918$m

= **11m** to two significant figures

(b) In reality, there will be some loss of energy in the swing.

(i) State one possible cause of energy loss.

Air/wind resistance and friction on swing ropes and the bar.

(ii) State how this energy loss could be reduced.

To reduce air resistance, streamline parts and to reduce friction, use lubrication on moving parts.